



**FACULTY OF SCIENCE, ENGINEERING
AND COMPUTING**

School of Computer Science and Mathematics

**BSc (Hons) DEGREE
IN**

Cyber Security and Digital Forensics

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"How do we decide what devices we provide?"

18th July 2024

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Declaration

I have read and understood the University regulations on plagiarism, and I understand the meaning of the word *plagiarism*. I declare that this report is entirely my own work. Any other sources are duly acknowledged and referenced according to the requirements of the School of Computer Science and Mathematics. All verbatim citations are indicated by double quotation marks ("..."). Neither in part nor in its entirety have I made use of another student's work and pretended that it is my own. I have not asked anybody to contribute to this project in the form of code, text, or drawings. I did not allow and will not allow anyone to copy my work with the intention of presenting it as their own work.

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Chapter 1: Introduction and State of the Art Review

1.1: Document Structure

This project report begins by exploring the decision-making process behind the provision of technological equipment (Tablets, Laptops, PCs) by governments, schools, employers and public buying considerations. The document is structured to provide a comprehensive overview of the criteria and considerations involved in selecting these devices, focusing on security, practicality, and reliability.

Chapter 1: Introduction and State of the Art Review

This chapter introduces the research question and context, offering a state-of-the-art review of current practices and standards in device provisioning.

Chapter 2: Criticism of Current Efforts and Standards

This section critically analyses existing efforts and standards in device provisioning, with specific sections examining measures taken by the UK and comparing these with practices in the United States. This chapter serves as a literature review, detailing the strengths and weaknesses of current approaches.

Chapter 3: Constructing Guidelines

This chapter outlines the guidelines for hardware and software considerations in device provisioning, focusing on user-friendliness and security measures. It includes tables of recommended software for various use cases (personal, student, work, and government) and discusses fundamental security considerations.

Chapter 4: Solution - Device Provisioning Application Overview

This chapter presents an overview of the proposed solution, a Device Provisioning Toolkit application. It describes the construction and features of the application, comparing it to competitors and concluding with recommendations on how to decide what devices to provide.

The references and bibliography section at the end of the document will provide a comprehensive list of all sources cited throughout the report, formatted in Harvard style.

1.2: Introduction

The aim of this dissertation is to explore the decision-making process behind the provision of technological equipment (tablets, laptops, PCs) by Governments, Schools, and Employers. The increasing reliance on technology in various sectors has made the provision of secure, practical, and reliable devices a critical issue. This dissertation addresses the research question: "How do we decide what devices we provide?" with a deeper look into, are we doing enough to protect them?

Many Companies, Governments/Councils and Schools pair with helpdesks (like Dataspire, n.d.), Crown Commercial Service Suppliers which are UK organisations acting as a central purchasing body for the government and public sector (www.crowncommercial.gov.uk, n.d.), Manufacturers (Microsoft Devices for Education) to get the support they need with buying devices to provision to employees and students. Organisations working in capacity with the Government such as Modis International LTD provide an end-to-end solution, much like

Crown Commercial Service Suppliers and Helpdesks, which provides expertise and support to organisations across HM Government and the wider public sector (www.applytosupply.digitalmarketplace.service.gov.uk, n.d.) offering the most support, guidelines, assistance that could be found, and aside from such companies, these providers do not supply any standards, guidelines or re-assurance that the devices offered have been specifically chosen with hardware and software security measures in mind, except that which is the governments subpar standard.

The vast majority of consumers utilize YouTube tech videos, sites & stores like Currys that provide online and instore meetings with store staff on which devices they should buy, online comparison sites and magazines (Passingham, 2020), or word of mouth and brand association. When looking at a device to buy, consumers largely rely on the brand for authenticity, the size for accessibility, the storage for capacity and the operating system for familiarity. When 1200 respondents were asked ‘How many reviews do you typically read before you make a decision to purchase?’, nearly 70% of respondents claimed to read between one and six customer reviews before making a purchasing decision (Pasquali, 2023). A huge reason for this is due to consumers wanting the best products they can afford, but not knowing how to go about ensuring that decision. One of the biggest issues with this is, aside from, online shopping scams accounting for ‘38 percent of all reported scams worldwide’ (cycles and Text, 2023), is the number of fake reviews online accounting for 4% of online reviews on average (World Economic Forum, n.d.). When put in context, an analysis by the fraudulent-review-detection service Fakespot found that ‘of 720 million Amazon reviews in 2020 about 42 percent were bogus.’ (www.ngpf.org, n.d.). This outlines a huge issue for consumers, the public and teams trying to buy devices, without the knowledge themselves on what makes a secure, reliable and practical device and purchase. Furthermore, it proves that Cybersecurity, is very rarely, considered if at all in this process.

Digital Literacy in the UK is heavily underdeveloped. ‘One in 10 adults has never used the internet’ (Gov.UK, 2017) The safety of a company and school needs to equally involve the safety precautions and knowledge of its employees and students. Company’s may do a good job in configuring monitoring systems, internet and operating system level restrictions, and User Access Control (UAC) permissions. However just as fire drills and safety training are important and mandatory, so is the digital security measures an employee should be taking. As pointed out by shrm.org, ‘Each employee provided with a laptop by [Company Name] is responsible for the physical security of the laptop. All laptops acquired for or on behalf of [Company Name] are deemed to be company property.’ (www.shrm.org, n.d.). The NCSC outlines a system that is Secure by Default needs to have security embedded within from the ground up (www.ncsc.gov.uk, n.d.). This should apply to a school, a college, a workplace, a government. Where each member and device are equipped with the resources to ensure security such as education, training, reliable and secure hardware considerations.

The department of education (DfE) has released minimum requirements and standards for Laptops, Desktops and Tablets for schools and colleges, for Laptops and Personal Computers being, but not limited to, Enterprise or education-grade operating systems, 5 years of support and security patches, 3 years of warranty and the device being recommended to be capable of meeting Wi-Fi 6 (802.11ax) standard. (Department of Education, 2024)

Between 2020 and 2022, the UK government distributed over ‘1.35 million laptops and tablets to schools, trusts, local authorities, and further education providers for disadvantaged children and young people’ as part of the Get Help with Technology (GHwT) program. (Education & Skills Funding Agency, 2022) Of these devices, guidance was provided for setting up devices such as Microsoft Windows Laptops and Tablets, Google Chromebooks, Apple iPads as well as a google domain for care leavers. (Department for Education, 2022) Optionally, devices ordered before October 2021 could have come standard or pre-installed with safeguarding software such as Cisco Umbrella and Microsoft Intune, but devices ordered later, ‘were delivered with factory settings and do not have any DfE-managed software installed.’ (Department for Education, 2020)

The guidance for securing these devices entailed steps from checking security features provided by the manufacturer were active and trusting them (Department for Education, 2020), keeping the devices updated ‘Chromebooks have the H1 Secure Microcontroller, which is Google’s version of TPM (Microsoft’s encrypted processor). This allows patching and OS updates to be installed more quickly, so it’s easier to keep devices up to date.’ (Department for Education, 2020) and enrolling in Mobile Device Management solutions (MDM) such as Apple School Manager (ASM) which enable IT admins to monitor, manage and secure multiple devices remotely. (Department for Education, 2020)

It is important to note these are guidance papers intended for the attention of IT departments within education or at the very least, a tech-savvy administrator with basic knowledge surrounding the tools and systems provided. This is made clear from the following quotes ‘When you ordered iPads, we asked for your Apple School Manager (ASM).’ (Department for Education, 2020), ‘The person who placed the Chromebook order received an email from deal-fulfillment@google.com, including important information about device setup. At the time we suggested they forward this email to the technical contact responsible for preparing Chromebooks.’ (Department for Education, 2020) and lastly, ‘If you do not have an MDM solution, you should look into the benefits, costs and resource requirements to understand whether device management is appropriate before you make a decision.’ (Department for Education, 2020).

Chapter 2. A Criticism on current efforts and standards

2.1 - Measures taken by the UK

In a setting, such as a school or college where devices have been donated by a GHwT scheme and thus a Bring Your Own Device (BYOD) policy may not be enforced, security measures such as an MDM should be paramount, not an option (Gema Howell (NIST), 2023) .

Employers and Tech support teams should be equipped with resources to protect and secure the devices they are given and provisioning, such as being able to scale and secure an MDM for maximum security and efficiency (Rhee et al., 2013) or understanding the Mobile Threat Catalogue by NIST to aid in protecting devices being provisioned (pages.nist.gov, n.d.). This information and access to basic resources shouldn't be simply left to an IT team or organisation to decide and provide recommendations for students and employees (www.kingston.ac.uk, n.d.) nor be entrusted to hardware and software manufacturers with a nod of approval from the government declaring their safety.

While action towards this is now being put in place, with resources and a thorough checklist of standards to ensure the presence of, and upkeep of, security within educational technology (Department for Education, 2024), information regarding best practices are kept optional behind a Cyber Essentials government-backed certification. The pages otherwise provided are tedious at best, requiring hours of scrolling through pages to clarify an original concept. For example, under the drop down titled, 'Devices should meet or exceed the minimum requirements', a reader is told:

'All devices should also meet or exceed the minimum requirements set out in this standard.

This will make sure devices are:

safe and secure

stable and reliable

Not meeting these requirements could mean your devices:

negatively impact teaching and learning

are at risk of malware, ransomware and potential data breaches

are not value for money'

To grasp the meaning of a 'safe and secure' device, a reader must access six more pages of topics just to be provided with a list of configuration measures that should be 'consider[ed] [them] when reviewing device requirements', leaving the subject of a 'safe and secure' device in the hands of whoever is to be configuring these devices.

The minimum requirements and standards for Laptops, Tablets and Desktops for students alongside the related standards and Cyber Essentials cyber security standards neglect important hardware and software considerations when it comes to work. The guidance provided by the UK government ("OFFICIAL") within these sources fails to fully address vital measures of importance that extend beyond these primitive 'requirements' and 'recommendations.

Such important recommendations, without mention, include measures such as: opting for PCs with DDR5 Ram (released to public July 14th 2020) to prevent rowhammer attacks (MITRE Corporation, 2020) such as RamBleed (rambleed.com, n.d.), CPU recommendations to avoid devices with 'pre-2019 microprocessors that use branch prediction' vulnerable to Meltdown and Spectre (nvd.nist.gov, n.d.), The importance of offline backups alongside encrypted cloud storage instead of as an alternative (www.gov.uk, n.d.) ensuring operating system

updates to protect against ransomware attacks such as WannaCry (BBC, 2017), operating system hardening methods and protocols provided by Microsoft in enterprise, pro and education editions, but do not come pre-configured as by default (support.microsoft.com, n.d.) relying on changes to a Group Policy set by an organisation, with no standards or regulations on doing so (Microsoft.com, 2024).

In addition to this, many basic considerations which should see fit as standard requirements for devices to ensure practicality, long lasting usability and efficiency of devices for students, employees and organisations to truly meet ‘digital and technology standards in schools and colleges’ instead of leaving the guess work to schools and colleges to decide their own standards are missing.

The department for work and pensions (DWP) has published an Acceptable Use Policy (DWP Acceptable Use Policy, n.d.) alongside a Remote Working Security Policy (DWP remote working security policy 1. Overview, n.d.) for its departments staff. Where expected considerations might include the use of a government virtual private network (VPN), proxy or official software for communication, privacy and security, advice for securing a home network, encrypting or password protecting files, readers are instead provided with a block list of sites and apps staff are not to access, being told to protect usernames, staff numbers, smart cards, dongles and passwords appropriately without mention of methods how to do so such as through password managers, and guidance on behavioural conduct to mitigate security issues. The Acceptable Use Policy does however indicate explicitly that, *‘Only use systems, applications, software and devices (including USBs, laptops and smart phones), which are approved, procured and configuration managed by DWP when undertaking official business, and apply DWP standards and guidance in their use. The installation and use of the TikTok application is restricted on all DWP devices and access must not be attempted.’* (8.1 Devices, Systems and Networks). From which we can infer there is in-depth IT standards and recommendations in place for the provisioning of, and maintenance of staff machines which is being handled by OFFICIAL IT departments respectively. However, the information as to what this might include is not immediately apparent.

The UK Government Technological Community listed among the gov.uk service manual exists for those ‘choosing technology (e.g. languages, frameworks or software) for a service, giving technology advice and being the technical voice in conversations about service design, making a service secure, testing the security and performance of a service,’ (www.gov.uk, 2018), but among the technology standards and guidance provided for Designing, building and buying technology for government, the guidance provided can be summarised to this statement from the source ‘Every service is different, and many of the decisions you make about technology will be unique to your service.’ (www.gov.uk, 2016). A statement which, coupled with a document that does not provide any recommendations or requirements in terms of hardware or software specifics, does not pay respect to the opening statement of this same source ‘Choosing technology is often the most significant area of investment you’ll make when developing or maintaining your service.’

The National Cyber Security Centre (NCSC) have provided a plethora of tools to assist Employees, Volunteers, Trustees, University research assistants, Students, MSP administrators, Third party contractors, Customers (National Cyber Security Centre, 2023) and Organisations with the task of readying, preparing and securing devices they have bought to a high level of security standards in methods easy to understand and put into practice. Available to access via the ukncsc Device Security Guidance Configuration Packs GitHub repository (GitHub, 2024) or via the Device Security Guidance site (www.ncsc.gov.uk, n.d.).

HM Government’s Government Functional Standard GovS 007: Security document (HM Government, 2021) outlines and covers in-depth security practices concerning ‘Physical Security’, ‘Cyber Security’, ‘Industry Security’, ‘Information Management’, ‘Capability,

capacity and resources', 'Personnel Security', 'Technical Security', 'Security Risk Management', 'Critical Assets and Resources', 'Security Culture, Education and Awareness' for government staff, organisations, departments and contractors.

2.2 - Areas to improve in comparison with the United States

None of these documents contain specifications for hardware or software, operating systems or important necessary security measures for all whether in collaboration with OFFICIAL or any organisation. Fundamental safety measures such as password protecting a compressed folder or file, using a vpn or guidelines for a safe, reliable and efficient are non-existent and pales in comparison to how the United States of America, for example, ensure the general public, students, employees, organisations and staff within OFFICIAL aren't just educated on Cyber Security standards, but are provided with guidance and specifications when it comes to deciding what they provide.

The United States maintain federal laws to ensure cybersecurity standards are met for the government, tribal governments, US workforce, education and public awareness (Drexel University Information Technology, 2021). These measures include, but aren't limited to, "Centralised management of software licenses", the use of 'virtual or cloud-based environments' and physical and virtual system monitoring for government devices to ensure the validity, authority and security of government-wide systems (INFORMATION TECHNOLOGY LAWS SECTION, n.d.). "Government workers at all levels are guided by policies, many of them changing at any given time to accommodate new rulings and legislation. Because of this dynamic environment, the availability of customization options is important when selecting government laptops." (MooringTech, n.d.).

The K-12 Cybersecurity Act of 2021 S.1917 'requires the Cybersecurity and Infrastructure Security Agency (CISA) to study the cybersecurity risks facing elementary and secondary schools and develop recommendations that include cybersecurity guidelines designed to assist schools in facing those risks.' (Peters, 2021). This bill ensures the education industry is provided with all the tools, standards, methods and knowledge necessary for building, protecting and monitoring their infrastructure. The bill provides school officials with publicly accessible cybersecurity guidelines and an online training toolkit.

US Government departments have outlined specifications for devices they require, but these have leeway, being more so guidelines for best practice and minimum expectations, 'The purpose of standard configurations and specifications is to define minimum required capabilities while also providing flexibility and allowing market competition on solutions within each category, driving value for the customer. The specifications in each category should be followed as closely as possible.' (Category Management Policy 15-1: Improving the Acquisition and Management of Common Information Technology: Laptops and Desktops Technical FAQs Configurations, n.d.). This document serves to support vendors provisioning government standard systems through the US Acquisition Gateway to meet minimum requirements government-wide. This means that each government department not only needs to provide hardware, software and security specifications are delivered, but also needs to ensure the guidance provided is up to date with current standards (www.sewp.nasa.gov, n.d.).

Workforce Innovation and Opportunity Act integrates technology into instruction, increasing student access to technology, and leveraging learning outside the classroom can help students develop the skills they need to support their long-term educational goals (World Education, n.d.).

This project not only aims to bring to light the important considerations missing from the guidelines and recommendations set by the government, organisations and the public, but to demonstrate an ideal tool to facilitate the provisioning of, and searching for, devices that not only conform, but perform to ideal standards, for the security and peace of mind for society.

Chapter 3. Constructing Guidelines

Many Crown Commercial Suppliers and ecommerce sites offering devices for Education (www.stonegroup.co.uk, n.d.), Personal Use, Government use and have filters and provide specifications for the devices they provide (LaptopsDirectUK, n.d.). Often the devices being provided however are not limited to only those under regulations, while it might be a good idea to facilitate those in charge with provisioning devices with all available stock. It is not necessary nor helpful. An ideal application should follow the following guidelines.

3.1 - Hardware & User Friendliness considerations

Sources online offer numerous specifications for recommended anti-virus and management solutions, best practices for securing devices, user friendly features and recommended hardware and software for both and personal use. While access to government practices has proven limited, service providers, such as Modis International LTD, offer vital assistance and expertise in configuring and securing company devices. Recently circling in popular media is of an official government standard operating system, namely Windows 11 G, a variant of windows 'pre-configured to remove features that are not needed by Chinese government employees' (TECHCOMMUNITY.MICROSOFT.COM, n.d.) this operating system is unofficial and false (Brinkmann, 2024), and while debloating Windows in an attempt might make the operating system itself secure, an operating system design like the proposed fake shows that such importance need be considered. Windows 365 Government edition does this (www.microsoft.com, n.d.), while complying to US Government Standards, it proves to be a vital step towards national security and industry standard that the UK should investigate developing themselves.

User friendliness should be considered, with ease to access and use the service this project aims to produce, and devices that perform in a secure, efficient manner, this may mean recommended Hardware, Software and peace of mind that fundamental security measures have been taken place, but also usability, lifetime of the device, familiarity, weight, size, operating system play a significant role in User Friendliness. Providing clear, user-friendly instructions for device provisioning and maintenance is crucial to ensure that users can easily manage and secure their devices.

In aiming to compile and develop hardware standards for UK devices as well as providing comfort for the user, in both for Work Use and Personal Use, various UK and US consumer and government sources have been compiled to clarify and ideal and agreed upon guideline.

Table of Device Recommendations for public by consumer sites

Device Type	Use Case	Model	OS	Processor	RAM	Storage	Screen Size & Resolution	Battery Life	Source
Laptop	Work	Acer Swift Edge	Windows 11 Home	AMD Ryzen 7 7840U	16GB	1TB SSD	16" 3.2K (3200 x 1800)	7.5 hours	Evening Standard, 2024 (Evening Standard) (Pritchard, 2024)
Laptop	Work	Dell XPS 13	Windows 11	Intel Core i5 (12th Gen)	8GB	256GB SSD	13.4" FHD+ (1920 x 1200)	8.5 hours	Independent, 2024 (The Independent) (The Independent, 2024)
Laptop	Work	HP Spectre x360	Windows 11	Intel Core i7	16GB	512GB SSD	13.5" FHD (1920 x 1080)	10 hours	TechRadar, 2024 (Whats The Best) (Whats The Best, 2024)
Laptop	Personal	MSI Modern 14	Windows 11	Intel i5-1235U	8GB	512GB SSD	14" FHD (1920 x 1080)	9 hours	Whatsthebest.co.uk , 2024 (Whats The Best) (Whats The Best, 2024)
Laptop	Personal	Lenovo IdeaPad 3i	Windows 11	Intel Pentium 8505	4GB	128GB SSD	17.3" FHD (1920 x 1080)	7.5 hours	Whatsthebest.co.uk , 2024 (Whats The Best) (Whats The Best, 2024)
Laptop	Student	ASUS Vivobook 16	Windows 11	Ryzen 7	8GB	512GB SSD	16" FHD (1920 x 1080)	7 hours	Whatsthebest.co.uk , 2024 (Whats The Best) (Whats The Best, 2024)
Laptop	Student	MacBook Air 13 (2017)	macOS	Intel Core i5	8GB	128GB SSD	13.3" Retina (2560 x 1600)	12 hours	Evening Standard, 2024 (Evening Standard) (Pritchard, 2024)
PC	Work	Dell RTX PC Ryzen 5	Windows 11 Pro	Ryzen 5	16GB	256GB SSD	15.6" FHD (1920 x 1080)	N/A	Yours.co.uk, 2024 (Yours) (Yours, 2024)
PC	Personal	Lenovo IdeaCentre AIO 3	Windows 10 Home	AMD Ryzen 3	8GB	1TB HDD + 256GB SSD	23.8" FHD (1920 x 1080)	N/A	Warwick IT , 2023, (warwick.ac.uk, n.d.)
Laptop / PC	Work	N/A	N/A	Intel Core i3 / M3 OR AMD Ryzen 3	4GB	128 GB SSD	Full HD (1080p) screen	N/A	Which 2020 (Passingham, 2020)
Laptop / PC	Work	N/A	N/A	Intel Core i5 / AMD Ryzen 5	8GB	256GB SSD	Full HD (1080p) screen	N/A	Which 2020 (Passingham, 2020)
Tablet	Personal	Samsung Galaxy Tab S8+	Android 12	Qualcomm Snapdragon 8 Gen 1	8GB	256GB SSD	12.4" Super AMOLED (2800 x 1752)	14 hours	Evening Standard, 2024 (Evening Standard) (Pritchard, 2024)

Table comparing device recommendations from UK sources

Table of Device Recommendations for employees and government by consumer sites

Device Type	Use Case	Processor	RAM	Storage	Display	Battery Life	OS	Additional Features	Source
Laptop	Business	Snapdragon X Elite (X1E-78)	16GB	512GB SSD	14" 2240x1400 (2.2K), touch, 300 nits	10+ hours	Windows 11 Pro	HP Wolf Pro Security, durable design	Windows Central, 2024 (Windows Central) (Bowden and updated, 2024)
Laptop	Consumer	Intel Core i7	16GB	1TB SSD	15.6" OLED, 4K	8-10 hours	Windows 11	High refresh rate, thin bezels	PCMag, 2024 (Beebom) (Sha, 2024)
Desktop PC	Business	Intel Core i9	32GB	2TB SSD + 2TB HDD	-	-	Windows 11 Pro	High performance, secure configurations	GSA, 2024 (U.S. General Services Administration) (U.S. General Services Administration, n.d.)
Desktop PC	Consumer	AMD Ryzen 9	16GB	1TB SSD	-	-	Windows 11	RGB lighting, customizable setup	Beebom, 2024 (Beebom) (Sha, 2024)
Tablet	Educational	MediaTek 8186	8GB	128GB SSD	10.1" 1920x1200, durable	12 hours	ChromeOS	Military-grade durability, impact shield	Laptop Mag, 2024 (LaptopMag) (published, 2024)
Laptop	Government	Intel Core i5	8GB	256GB SSD	13.3" 1920x1080	8+ hours	Windows 11	Secure-core PC, long battery life	GSA, 2024 (U.S. General Services Administration) (U.S. General Services Administration, n.d.)
Tablet	Consumer	Qualcomm Snapdragon 888	6GB	128GB SSD	12.4" Super AMOLED, 2800x1752	10+ hours	Android	S Pen support, high brightness	Digital Trends, 2024 (Windows Central) (Bowden and updated, 2024)
Laptop	Consumer	AMD Ryzen 5	8GB	512GB SSD	14" 1920x1080, IPS	7-8 hours	Windows 11	Value for money, portable	Windows Central, 2024 (Windows Central) (Bowden and updated, 2024)

Table comparing device recommendations from US Consumer & Government sources

The GSA provides detailed guidelines for purchasing laptops and desktops for federal use. These standards emphasize secure configurations, durability, and long battery life. They recommend using processors such as Intel Core i5 and higher, with at least 8GB of RAM and SSD storage to ensure performance and security for government applications.

Educational standards in the UK emphasize the need for durable and secure devices that can handle the rigorous use in schools. Devices recommended for educational use often include rugged tablets with long battery life and enterprise-grade security features. The use of ChromeOS and managed configurations is also common to ensure ease of maintenance and security compliance.

Consumer-oriented sources like Whatsthebest and Which focus on the latest advancements in consumer technology, such as high-resolution displays, powerful processors like Intel Core

i7, and extensive battery life. These reviews often highlight features like high refresh rates, OLED displays, and advanced security features.

Based on this compiled list the following recommendations can be formed.

Work Use

Laptops

Processor: Intel Core i5/i7 (12th Gen) or AMD Ryzen 7
Sources: The Independent, TechRadar, Evening Standard
RAM: 16GB
Sources: Evening Standard, Warwick IT Services
Storage: 512GB SSD or 1TB SSD
Sources: PCMag, Windows Central
Display: 13.3" to 16" with FHD+ or higher resolution
Sources: Independent, TechRadar
Battery Life: 7-10 hours
Sources: Evening Standard, Independent

Desktops

Processor: Intel Core i7/i9 or AMD Ryzen 5/9
Sources: GSA, Beebom
RAM: 16GB or 32GB
Sources: GSA, Warwick IT Services
Storage: 1TB SSD or 2TB SSD + 2TB HDD
Sources: GSA, Warwick IT Services

Personal Use

Laptops

Processor: Intel Core i5 or AMD Ryzen 5

Sources: Whatsthebest.co.uk, Windows Central

RAM: 8GB

Sources: Whatsthebest.co.uk, Windows Central

Storage: 256GB SSD or 512GB SSD

Sources: Whatsthebest.co.uk, Digital Trends

Display: 14" to 15.6" with FHD resolution

Sources: Whatsthebest.co.uk, PCMag

Battery Life: 7-9 hours

Sources: Windows Central, Evening Standard

Desktops

Processor: AMD Ryzen 5 or Intel Core i5/i7

Sources: Beebom, Warwick IT Services

RAM: 8GB or 16GB

Sources: Beebom, Warwick IT Services

Storage: 512GB SSD or 1TB HDD + 256GB SSD

Sources: Beebom, Warwick IT Services

Student Use

Laptops

Processor: AMD Ryzen 7 or Intel Core i5

Sources: Whatsthebest.co.uk, Evening Standard

RAM: 8GB

Sources: Whatsthebest.co.uk, Warwick IT Services

Storage: 256GB SSD or 512GB SSD

Sources: Whatsthebest.co.uk, Evening Standard

Display: 13.3" to 16" with FHD resolution

Sources: Whatsthebest.co.uk, Independent

Battery Life: 7-12 hours

Sources: Evening Standard, Independent

Government Use

Laptops

Processor: Intel Core i5 or higher

Sources: GSA, Warwick IT Services

RAM: 8GB

Sources: GSA, Warwick IT Services

Storage: 256GB SSD

Sources: GSA, Warwick IT Services

Display: 13.3" with FHD resolution

Sources: GSA, Warwick IT Services

Battery Life: 8+ hours

Sources: GSA, Warwick IT Services

Summary

Work Use:

High performance with Intel Core i5/i7 or AMD Ryzen 7, 16GB RAM, 512GB SSD, FHD+ displays, and long battery life.

Personal Use:

Balanced performance with Intel Core i5 or AMD Ryzen 5, 8GB RAM, 256GB/512GB SSD, FHD displays, and moderate battery life.

Student Use:

Affordable yet capable with AMD Ryzen 7 or Intel Core i5, 8GB RAM, 256GB/512GB SSD, FHD displays, and extended battery life.

Government Use:

Secure and efficient with Intel Core i5, 8GB RAM, 256GB SSD, FHD displays, and reliable battery life.

Minimum recommendations being, Intel Core i5, 8GB Ram, 256 GB SSD, FHD Display and moderate reliable battery life.

These recommendations provide a comprehensive overview of optimal hardware configurations for different use cases, ensuring that devices meet specific requirements for performance, durability, and security across various sectors.

3.2 - Software Measures for consideration

In-depth guidance on a system basis would be vital to ensuring a device being bought was compatible and secure. Such as available / recommended software, MDM management solutions, Anti-Virus tools. Guidance should be applicable to the device and UK Governmental standards. Emphasizing the importance of selecting software that meets security and compatibility standards to maintain system integrity and protect against cyber threats. The following tables present a detailed overview of recommended security software for various use cases: personal, student, work, and government use. Each entry is supported by multiple sources.

Table of devices for Personal Use from consumer sources

Software	Description	Sources
Norton 360	Comprehensive protection suite	Windows Central, (updated, 2023)
McAfee Total Protection	Antivirus and identity protection	Digital Trends, (Digital Trends, 2024)
Bitdefender Total Security	Multi-layered security	Windows Central, Digital Trends, PCMag
Kaspersky	Antivirus and internet security	PCMag, Evening Standard, Whatsthebest.co.uk
Avast Free	Free version with strong protection and additional tools like a VPN.	Windows Central, (updated, 2021)
Avira Prime	All-in-one security suite	PCMag, Evening Standard, Digital Trends

For personal use, comprehensive security suites like Norton 360, McAfee Total Protection, and Bitdefender Total Security are highly recommended. These suites offer multi-layered protection, including antivirus, firewall, and identity theft protection, providing users with extensive security coverage. Sources such as PCMag, Digital Trends, and Windows Central consistently highlight these products for their robust features and user-friendly interfaces.

Table of devices for Student Use from consumer sources

Software	Description	Sources
Norton 360	Comprehensive protection suite	Whatsthebest.co.uk, PCMag, Digital Trends, Windows Central
McAfee Total Protection	Antivirus and identity protection	Evening Standard, Whatsthebest.co.uk, Digital Trends
Bitdefender Total Security	Multi-layered security	Windows Central, Digital Trends, PCMag
Kaspersky Total Security	Antivirus and internet security	PCMag, Evening Standard, Whatsthebest.co.uk
Avast Premium Security	Antivirus with extra features	Windows Central, Digital Trends, Whatsthebest.co.uk
Avira Prime	All-in-one security suite	PCMag, Evening Standard, Digital Trends

Students require reliable and affordable security solutions that can protect against common threats while ensuring system performance. Norton 360, McAfee Total Protection, and Bitdefender Total Security are again the top recommendations, given their comprehensive protection and ease of use. Educational discounts and multi-device licenses make these solutions particularly attractive for students.

Table of devices for Work Use from consumer sources

Software	Description	Sources
Symantec Endpoint Protection	Comprehensive endpoint security	Whatsthebest.co.uk, PCMag, Digital Trends, Windows Central
McAfee Endpoint Security	Business-focused antivirus	Evening Standard, Whatsthebest.co.uk, Digital Trends
Bitdefender GravityZone	Advanced threat protection	Windows Central, Digital Trends, PCMag
Kaspersky Endpoint Security	Robust security for businesses	PCMag, Evening Standard, Whatsthebest.co.uk
Sophos Intercept X	Endpoint protection with EDR	Windows Central, Digital Trends, Whatsthebest.co.uk
Trend Micro Apex One	Comprehensive endpoint protection	PCMag, Evening Standard, Digital Trends

Work environments demand advanced security solutions capable of protecting sensitive data and managing multiple endpoints. Symantec Endpoint Protection, McAfee Endpoint Security, and Bitdefender GravityZone are preferred choices for businesses due to their extensive feature sets, including advanced threat protection, centralized management, and compliance capabilities. Sources such as Whatsthebest.co.uk and PCMag emphasize the importance of robust endpoint security in protecting corporate networks.

Table of devices for Government Use from consumer sources

Software	Description	Sources
Symantec Endpoint Protection	Comprehensive endpoint security	Whatsthebest.co.uk, PCMag, Digital Trends, Windows Central
McAfee Endpoint Security	Business-focused antivirus	Evening Standard, Whatsthebest.co.uk, Digital Trends
Bitdefender GravityZone	Advanced threat protection	Windows Central, Digital Trends, PCMag
Kaspersky Endpoint Security	Robust security for businesses	PCMag, Evening Standard, Whatsthebest.co.uk
Sophos Intercept X	Endpoint protection with EDR	Windows Central, Digital Trends, Whatsthebest.co.uk
Trend Micro Apex One	Comprehensive endpoint protection	PCMag, Evening Standard, Digital Trends

Government use necessitates the highest level of security, given the sensitivity of data and the potential impact of breaches. Symantec Endpoint Protection, McAfee Endpoint Security, and Bitdefender GravityZone are frequently recommended for their comprehensive security measures, including encryption, advanced threat detection, and policy enforcement. These solutions are endorsed by various sources, highlighting their effectiveness in securing government devices.

The landscape of software aimed at securing devices such as laptops, PCs, and computers is robust and diverse, tailored to meet the specific needs of different user groups: personal, student, work, and government use. Norton 360, for instance, is highly recommended across multiple platforms like PCMag, DigitalTrends, and WindowsCentral due to its comprehensive protection features and user-friendly interface, making it suitable for personal and student use. McAfee Total Protection, also recognized by sources such as Standard and DigitalTrends, offers strong security measures ideal for personal and student use, with additional focus on ease of use and affordability.

Bitdefender Total Security stands out for its advanced threat defence mechanisms, with endorsements from WindowsCentral and PCMag highlighting its suitability for both personal and student users who require high levels of security without compromising on performance. For professional environments, Symantec Endpoint Protection and McAfee Endpoint Security are prominently recommended. These software solutions are noted for their robust enterprise-grade security features and are often highlighted in PCMag and DigitalTrends as top choices for work-related security needs.

In government use, where the importance of security is highest, solutions like Bitdefender GravityZone and Kaspersky Endpoint Security are frequently mentioned. Sources such as

WindowsCentral and PCMag praise these tools for their extensive security protocols and reliability in handling sensitive information.

Overall, the selection of security software is diverse, with each product offering unique features tailored to different user requirements. The consistent recommendations across reputable sources underscore the importance of choosing the right security software based on specific needs and usage contexts.

To provide a product that assists with device provisioning, latest statistics and recommendations regarding best security software should be provided to assist in the buying decision. Not simply discounts, low prices, complicated terms, but recommendations through proven measures provided.

The need for robust and comprehensive security software is evident. Norton, McAfee, Bitdefender, Kaspersky, Sophos, and Trend Micro consistently emerge as top recommendations due to their extensive features and proven effectiveness. With the ability to leverage the insights from multiple sources, users can make informed decisions to protect their devices and data effectively.

3.3 - Recommended Software & Security Considerations

Security considerations for provisioning devices should include mandatory links to toolkits for securing both the operating system and hardware. This involves using National Cyber Security Centre (NCSC) toolkits and best practices, such as encrypting BIOS and implementing hard drive backups. It's essential to provide comprehensive guidelines to ensure devices are safe from malware, ransomware, and data breaches. Ensuring the operating system is hardened and regularly updated is also vital. Security of these devices, whether kept protected at work, at home or in public, means considering measures that affect a wide range of areas, from hardware specifications, operating system measures, as well as physical security measures, networks being used etc... As aforementioned standards are very specifically outlined for physical security, but measures such a software to use or OS hardening methods lack presence. Such recommendations to be provided should include:

Table of Fundamental Security Recommendations for Devices

Measure	Device Type	Description	Source	URL	Quote
BitLocker Drive Encryption	Laptops, PCs	Encrypts entire volumes to protect data.	Microsoft Learn - BitLocker Overview	(Microsoft, 2023)	"BitLocker Drive Encryption provides offline data and operating system protection by ensuring that the drive is encrypted."
Windows Defender Application Guard	Laptops, PCs	Isolates untrusted sites and protects against malware.	Microsoft Learn - Application Guard	(vinaypamnani-msft, 2023)	"Windows Defender Application Guard helps protect your device from advanced attacks by isolating untrusted websites in a secure container."
Credential Guard	Laptops, PCs	Protects credentials by isolating them from the rest of the OS.	Microsoft Learn - Credential Guard	(paolomatarazzo, 2023)	"Credential Guard uses virtualization-based security to isolate secrets so that only privileged system software can access them."
Device Guard	Laptops, PCs	Uses hardware and software to ensure only trusted applications can run.	Microsoft Learn - Device Guard	(jsuther1974, 2023)	"Device Guard is a combination of enterprise-related hardware and software security features that, when configured together, will lock a device down so that it can only run trusted applications."
Windows Hello for Business	Laptops, PCs	Multi-factor authentication including biometric security.	Microsoft Learn - Windows Hello for Business	(paolomatarazzo, 2023)	"Windows Hello for Business replaces passwords with strong two-factor authentication on PCs and mobile devices."
Enable Auto-Lock	Laptops, Tablets	Devices lock automatically after inactivity.	The Windows Club	(support.microsoft.com, n.d.)	"Auto Lock feature allows Windows to automatically lock after a period of inactivity to enhance security."
Secure Boot	Laptops, PCs	Ensures that the device boots	Microsoft Learn - Secure Boot	(windows-driver-content, 2021)	"Secure Boot helps prevent malicious

		only using software trusted by the manufacturer.			software and unauthorized operating systems from loading during the system start-up process."
VPN Usage	Laptops, Tablets	Ensures secure remote access via VPN on public networks.	The Windows Club	(support.microsoft.com, n.d.)	"Setting up a VPN in Windows 10 can provide a secure way to connect to your workplace or access your data remotely."
Group Policy Object (GPO)	PCs	Enforces security settings across networked devices.	The Windows Club	(Microsoft, 2018) link	"Group Policy Object (GPO) can be used to enforce security settings and configurations across all computers in a network."
Regular Patching	PCs	Keeps systems updated with the latest security patches.	The Windows Club	(blog.scalefusion.com, 2024)	"Regularly checking for and applying Windows updates can help protect your computer from security vulnerabilities."
Endpoint Protection	Laptops, PCs	Uses Windows Defender Antivirus for malware protection.	The Windows Club	(BalaDelli, n.d.)	"Windows Defender Antivirus provides comprehensive, built-in protection against malware and other threats."
Application Whitelisting	Laptops, PCs	Restricts device execution to a set of approved applications.	Microsoft Learn - Application Control	(jsuther1974, 2023)	"Application whitelisting restricts which applications are allowed to run, thereby helping to protect systems from running malicious software."
Data Loss Prevention	Laptops, PCs	Prevents sensitive data from being shared outside the organization.	Microsoft Learn - Data Loss Prevention	(Microsoft, 2023)	"Data Loss Prevention (DLP) helps to detect and prevent sensitive information from being shared outside your organization."
Least Privilege Principle	Laptops, PCs	Ensures users only have access necessary for their roles.	Paloaltonetworks - What Is the Principle of Least Privilege?	(Palo Alto Networks, n.d.)	The principle of least privilege ensures users have only the access they need to perform their job functions.
Network Segmentation	PCs	Isolates different parts of the network to limit access and improve security.	CISCO - <i>What Is Network Segmentation</i>	(Cisco, 2019)	"Segmentation divides a computer network into smaller parts. The purpose is to improve network performance and security. Other terms that often mean the same thing are network segregation, network partitioning, and network isolation."

Multi-Factor Authentication (MFA)	Laptops, PCs	Requires multiple forms of verification for access.	Microsoft Learn – What is Multifactor Authentication	(Microsoft, n.d.)	"Multi-factor authentication (MFA) adds an extra layer of security by requiring multiple forms of verification to access resources."
Disable Unnecessary Services	Laptops, PCs	Reduces attack surface by disabling unused services.	UC SANTA CRUZ – Turn off unnecessary services	(Ucsc.edu, 2015)	Disabling unnecessary services can help reduce the attack surface and improve the overall security of the system.
OS Hardening	Laptops, PCs	Involves securing the operating system by applying security patches and configurations.	Perception Point OS Hardening 10 Best Practices	(Perception Point, n.d.)	"OS hardening involves applying security configurations and updates to minimize vulnerabilities."
User Training and Awareness	All Devices	Educates users on security best practices and threat awareness.	Barracuda - What is User Security Awareness?	(Barracuda Networks, n.d.)	"User security awareness training provides employees with the information they need to understand the dangers of social engineering, detect potential attacks, and take the appropriate actions to protect your business with security best practices."
Regular Backups	PCs	Ensures data is regularly backed up to recover from data loss.	UK Data Service - Backup	(Service, n.d.)	"Regular backups help protect against accidental or malicious data loss due to: human error hardware failure software or media faults virus infection or malicious hacking power failure"

Alongside these Security Measures the application database aims to ensure all devices are not vulnerable to meltdown and spectre CPU attacks, Rowhammer and other hardware attacks using updated lists and ensuring the devices provided are upto date, transparency being a key feature for consumers.

Chapter 4 – Solution: Device Provisioning Application Overview

4.1 - Constructing the Application

With these necessities considered a suitable application, such as the Device Provisioning Toolkit created by this project, will follow the following MoSCoW requirements.

Must Have

Security Measures:

- Detailed flowchart creation that includes security checks.
- Integration of current security standards and protocols.
- Recommendations for hardware using the standards outlined (e.g., DDR5 RAM, Intel Core i5, 8GB Ram, 256 GB SSD, FHD Display and moderate reliable battery life) and software (e.g., MDM solutions, Antivirus tools).
- Steps for operating system hardening and update protocols.

User Guidance:

- Clear, user-friendly instructions for device provisioning.
- Links to resources and toolkits for securing devices.
- Guidelines for selecting practical, reliable, and secure devices.

Comprehensive Standards:

- Adherence to government and industry standards.
- In-depth recommendations for hardware and software specifications.
- Incorporation of Cyber Essentials guidelines and other relevant standards.

Educational Resources:

- Training modules for digital literacy.
- Resources for understanding cybersecurity and best practices.

Scalability:

- Ability to scale recommendations based on the size and needs of the organization (e.g., schools, companies, government departments).

Should Have

Compatibility Checks:

- System compatibility verification tools for hardware and software.
- Recommendations for operating systems and applications suitable for different user needs.

Review and Feedback Mechanism:

- Option for users to provide feedback on device performance and security.
- Incorporation of user reviews and ratings to guide future users.

Device Management Tools:

- Integration with Mobile Device Management (MDM) solutions.
- Monitoring and management capabilities for IT administrators.

Could Have

Customization Options:

- Customizable recommendations based on specific organizational needs.
- Advanced settings for tech-savvy users to tweak recommendations.

Offline Capabilities:

- Functionality for offline access to guides and toolkits.
- Backup and encryption recommendations for offline use.

Interactive Elements:

- Interactive tutorials and videos for device setup and security.
- Virtual assistant for real-time support and guidance.

Won't Have

Non-Standard Devices:

- Recommendations for non-standard or experimental hardware and software.
- Support for devices outside the scope of government and industry standards.

Unverified Sources:

- Incorporation of unverified user reviews or non-validated security measures.
- Endorsement of third-party software without thorough vetting.

4.2 - Application Overview

To construct this application in a way that was comfortable, easy to use, customisable and reliable, it was built with a python flask web-app structure in mind. This allowed progressing on developing and future maintaining of the application to be done in a scalable and efficient way. This application takes into considerations all the measures outlined in this document and brings it all together comfortably in a simple manner. With transparency concerning device specifics, supplying the consumer with only secure and specified goods (laptop, pc and tablets), and flowchart creation for a chosen device to assist buyers in following a set of minimum requirements concerning safety measures that need be taken to secure their devices efficiently.

This app aims to pose as the best site to go to find, ensure the safety of, buy and manage devices for all security conscious users.

Images of the application with explanations:

Image 1: Landing Page with Recommended Devices

Below the search form, a carousel displays recommended devices. Users can scroll through the carousel to view different devices. Each device card includes an image, name, category, CPU speed, RAM, storage, screen size, and price.

Security Recommendations

The security recommendations section provides guidelines to ensure the devices are secure and up-to-date. It includes links to resources such as Cyber Essentials Guidelines, Operating System Hardening Steps, and Recommended Antivirus Tools.

Educational Resources

The educational resources section links to modules on digital literacy and cybersecurity.

It aims to help users enhance their understanding of device security and best practices.

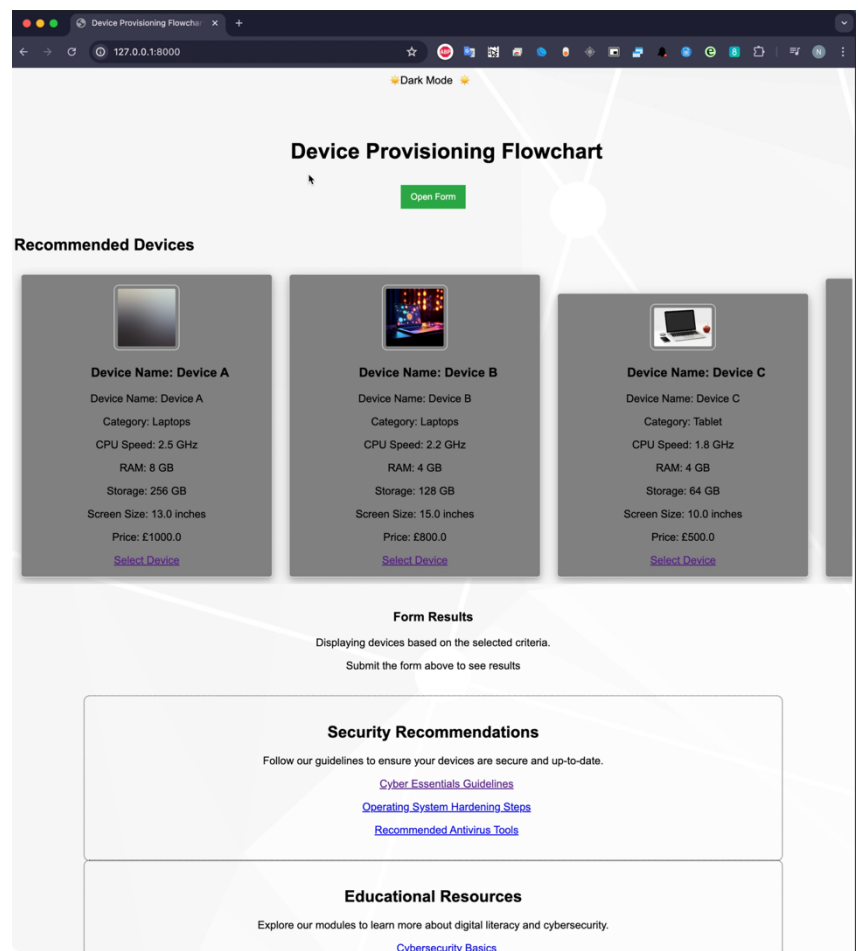


Image 2: Submit and Search Form

The home page features an interactive form where users can input a specific name for the device and drag the sliders to see listings within their price range.

Expanding Additional Filters allows for further customisation to their device search criteria, such as, usage type, device type, CPU speed, RAM storage, storage, and screen size.

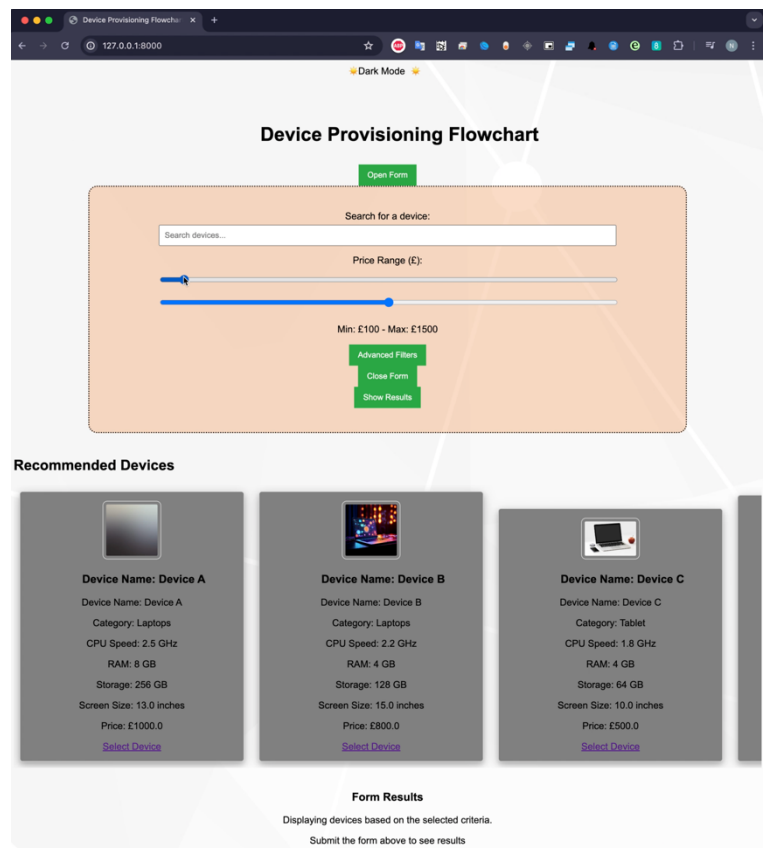


Image 3: Form Results

After submitting the search form, the results are displayed below the recommended devices section. The results show devices that match the user's specified criteria. Each result card provides detailed information about the device, a preview image for the device and a link to view more details.

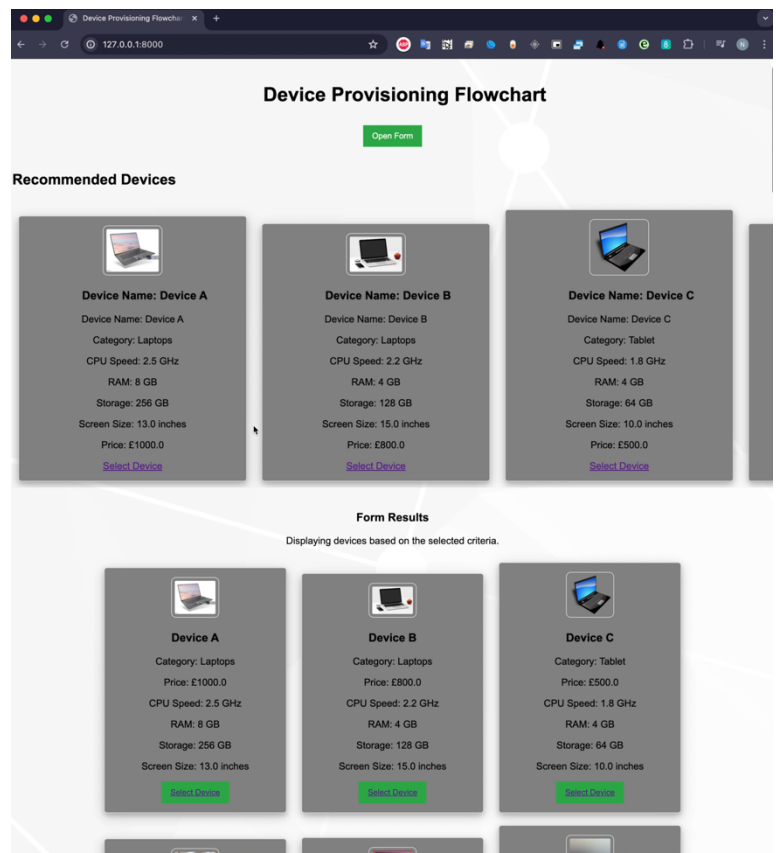
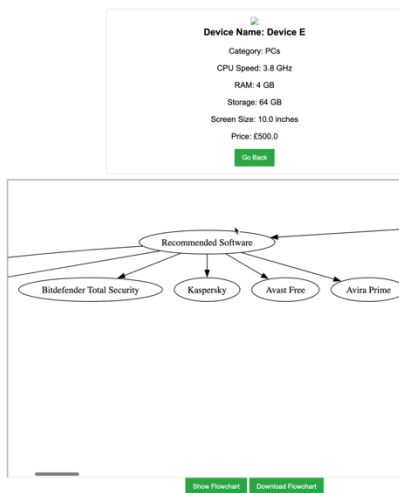
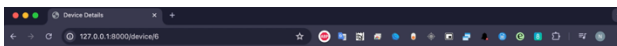
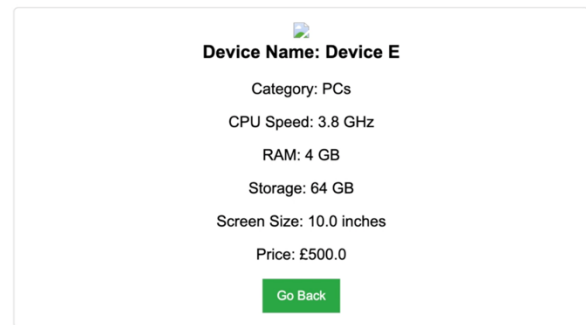
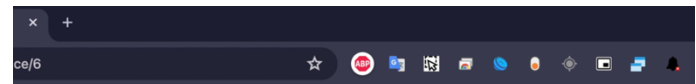


Image 4: Device Details Page

When a user clicks on a device card, they are taken to a detailed device page. This page displays all relevant information about the selected device and includes a flowchart that guides the user on security measures and recommended software for the device.



4.3 - Competitors for public and private sector device provisioning

While there is no single competitor providing such a product, putting this tool in its own market, it is not completely unique.

There is however a competitive landscape for providing secure and practical devices to the public and private sectors. With each competitor presenting unique strengths and facing distinct challenges, the SWOT analysis reveals the critical factors shaping this market, emphasizing the need for strategic approaches in decision-making. Such factors include improving transparency between each device's security features, providing toolkits, recommended software and hardware, knowledge for setup and configuration as well as a wide range of devices that meet government and consumer standards to ensure performance, practicality and security.

Strengths such as robust brand reputations, innovative technologies, and extensive support services distinguish companies like Apple, Microsoft, and Dell.

On the other hand, specialized vendors like ExCotek and Mooring Tech excel in tailored solutions for government and educational institutions.

Weaknesses identified across these competitors include high costs, supply chain dependencies, limited customization options, lack of fundamental security features or considerations which can hinder their ability to meet diverse client needs and don't necessarily ensure a device is practical or secure.

Opportunities are abundant in this sector, driven by increasing demand for secure devices and the expansion of cloud services and hybrid technologies. The public sector, particularly through entities like Crown Commercial Services and various US government sellers, presents a substantial market for secure technology solutions. However, these opportunities come with threats such as intense market competition, evolving regulatory landscapes, and economic instability that can impact growth and profitability.

To provide a service that meets the fundamental needs consumers currently lack a strong focus on innovation, security, and compliance to meet the evolving demands of both consumers and government entities needs to be present.

In conclusion, this analysis underscores the importance of a strategic, well-rounded approach to navigating the competitive landscape. By continually innovating and adapting to market changes, companies can secure their position and drive success in providing secure and practical devices for diverse sectors.

The SWOT analysis table below competitors that are used by consumers in both public and private sector.

SWOT analysis of Competitors:

Competitor	Strengths	Weaknesses	Opportunities	Threats
Apple	Strong brand reputation, high-quality products, loyal customer base, innovative technology	High prices, limited product customization	Expansion into new markets, growing demand for premium devices	Intense competition, potential supply chain disruptions
Microsoft	Comprehensive product range, strong enterprise relationships, robust software integration	Perceived as less innovative in hardware, dependency on third-party manufacturers	Growing cloud services, increasing demand for hybrid devices	Competitive pressure from both hardware and software rivals
Google	Strong software ecosystem, integration with Google services, innovative features	Limited hardware portfolio, less focus on premium segment	Expansion in smart home and AI-driven devices, increasing Chromebook adoption	Privacy concerns, intense competition in both hardware and software
Dell	Strong enterprise solutions, wide product range, customizable devices	Less brand appeal among general consumers, dependency on enterprise market	Growth in remote working trends, increasing demand for high-performance PCs	Price competition, supply chain challenges
HP	Wide product range, strong global presence, good balance of price and performance	Perceived as less innovative, issues with product reliability	Expansion in emerging markets, increasing demand for sustainable products	Competitive market, price sensitivity among consumers
Lenovo	Strong market position, wide range of innovative products, competitive pricing	Dependence on supply chain stability, perceived as less premium	Growth in education and business sectors, expanding product lines	Geopolitical tensions, intense competition from other PC manufacturers
Acer	Competitive pricing, strong presence in education market, good value for money	Lower brand recognition in premium segment, product reliability issues	Growing demand for affordable devices, increasing presence in gaming laptops	Price wars, market saturation
Asus	Innovative gaming laptops, strong presence in consumer electronics, competitive pricing	Less focus on enterprise market, perceived as less reliable	Growth in gaming industry, increasing demand for high-performance consumer laptops	Intense competition in gaming laptops, fluctuations in component prices
Samsung	Strong brand, wide product range, innovative technology	Higher prices compared to some competitors, fragmented product strategy	Growth in smart home integration, increasing demand for high-quality displays	Market competition, potential supply chain disruptions
Toshiba (Dynabook)	Strong reputation in business laptops, good balance of price and performance, reliable products	Less focus on consumer market, limited innovation in recent years	Growth in business sectors, increasing demand for secure and reliable laptops	Intense competition, supply chain vulnerabilities
Amazon	Vast product selection, strong logistics network, customer reviews and ratings	Issues with counterfeit products, variable seller reliability	Expanding private label brands, growth in subscription services	Regulatory scrutiny, competition from other e-commerce platforms
Newegg	Strong focus on technology and electronics,	Limited global presence, less	Growth in niche tech markets, increasing	Intense competition from larger e-commerce

	comprehensive product reviews, competitive pricing	appealing user interface	demand for high-performance components	players, potential supply chain disruptions
Best Buy	Strong customer service, wide range of electronics, physical store presence	Higher prices compared to online-only retailers, limited global reach	Growth in smart home devices, increasing demand for in-store tech support	Competitive pressure from online retailers, changing consumer shopping habits
Currys	Strong UK presence, wide product range, in-store support and services	Limited international presence, perceived as less innovative	Growth in smart home and IoT devices, increasing demand for tech support services	Competition from online retailers, potential economic downturn impacting consumer spending
ExCotek (Crown Commercial Supplier example)	Specializes in secure and compliant technology solutions, strong public sector experience	Limited to UK market, potential bureaucratic delays	Increasing government demand for secure and compliant IT solutions	Budget constraints in public sector, competition from larger multinational companies
Mooring Tech (US Gov Supplier example)	Trusted provider for US government, expertise in secure technology solutions, strong support	Lengthy procurement processes, dependence on government contracts	Expansion into federal technology initiatives, growing need for secure and compliant devices	Budget constraints, regulatory changes, competition from other government-focused providers

4.4 - Conclusion – How should we decide what devices we provide?

This project aims to make a push for cyber security within our daily lives, removing cyber security from being an unbacked and undefined buzzword within news outlets, consumer sites and government documents and outlining steps and actions that need to be taken to put cyber security in the forefront of our decisions while giving light to other issues that need be addressed such as the needs of mandatory initiatives to raise the level of digital literacy, The supply to governments, organisations and educational industries of toolkits and more in-depth guidance on device provisioning security measures and an easy and reliable way to buy practical and secure devices for their needs regardless of their industry. For these measures, further research and work needs to be done to ensure that cyber security is taken seriously and put at the forefront of our minds in the UK. Such initiatives like Open Age courses provide OAPs with improving on and developing s fundamental digital literacy. The product produced by this project aims to do the same, improving the digital literacy within the UK as well as ensuring that consumers are kept informed and are provided trusted, practical and secure devices.

References

The dissertation will include a comprehensive list of references within a bibliography citing all the sources, studies, and materials used in the research. The presentation of the dissertation will adhere to academic standards, including proper formatting, citation style, and clear organization of content.

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